

METADATA

PROJECT PREFIX		PROCESS UNIT CODE TABLE	
Projet Prefix	Facility / Building Description	Unit Process Description	Unit Process Code
WDP00	Domestic Pumping Common	UTILITIES	000
WDP01	Domestic Used Water Coarse Screening Shaft	Fire Water	001
WDP12	Domestic Influent Pumping Station 1	Grade 1 Plant Water	002
WDP22	Domestic Influent Pumping Station 2	Grade 2 Plant Water	003
WDP03	iPS Odour Treatment Plant	Grade 3 Plant Water	004
WIPO0	Industrial Pumping Common	Potable Water	005
WIPO1	Industrial Used Water Coarse Screening Shaft	Instrument Air	006
WIP12	Industrial Influent Pumping Station	Plant Air	007
WDL00	Domestic Liquids Common	Plant Drainage	008
WDL11	Domestic LM1 Headworks	INFLUENT PUMPING	010
WDL21	Domestic LM2 Headworks	Coarse Screening	011
WDL12	Domestic LM1 Primary Treatment	Used Water Pumping	012
WDL22	Domestic LM2 Primary Treatment	Ancillaries	013
WDL13	Domestic LM1 Secondary Treatment	PRELIMINARY TREATMENT	020
WDL23	Domestic LM2 Secondary Treatment	Intermediate Screening	021
WDL15	Domestic LM1 NEWater Treatment	Fine Screening	022
WDL25	Domestic LM2 NEWater Treatment	Screenings Washing & Loadout	023
WDL06	Domestic Chemical Area	Grit Removal	024
WILO0	Industrial Liquids Common	Grit Classification & Loadout	025
WILO1	IUW Equalisation	Fats, Oils & Grease Removal	026
WIL11	Industrial LM1 Headworks	Load Equalisation	027
WIL21	Industrial LM2 Headworks	PRIMARY TREATMENT	030
WIL12	Industrial LM1 Primary Treatment	Rapid Mixing	031
WIL22	Industrial LM2 Primary Treatment	Flocculation	032
WIL13	Industrial LM1 Secondary Treatment	Biosorption	033
WIL23	Industrial LM2 Secondary Treatment	Lamella Settling	034
WIL04	ILM Odour Facility	Primary Sludge Pumping	035
WIL15	Industrial LM1 IW Treatment	Primary Sludge Recycle Pumping	036
WIL25	Industrial LM2 IW Treatment	Primary Scum Pumping	037
WIL06	Industrial Chemical Area	SECONDARY TREATMENT	040
WSB00	Biosolids Building Common	MBR Fine Screening	041
WSB01	Biosolids Building	Bioreactors	042
WSB12	Biosolids OCF	Aeration Blowers	043
WSD00	Digestion Common	Secondary Membranes	044
WSD11	Digestion Pre-Treatment 1	Membrane Tanks & MBR Filtrate Pumping	045
WSD21	Digestion Pre-Treatment 2 (Future)	Deoxygenation Tanks	046
WSD02	Digestion	RAS Pumping	047
WSD03	Biogas Storage	WAS Pumping and Separation	048
WSD04	Biogas Burner	WATER RECLAMATION	050
WSD05	Greasy Waste Receiving Facility	MBR Filtrate Storage & Transfer Pumping	051
WSD06	GWRf OCF	RO Feed Tanks	052
WSD07	Digester Feed Pumping Station	RO High Pressure Feed Pumping	053
WST00	Sidestream Treatment Common	Cartridge Filters	054
WST01	Sidestream Phosphorus Removal	Reverse Osmosis	055
WST02	Sidestream Nitrogen Removal	RO Flush Tank and CIP	056
WWW00	Wet Weather Treatment Facility Common	UV Disinfection	057
WWW11	Wet Weather Treatment Facility 1	CO2 Stripping	058
WWW21	Wet Weather Treatment Facility 2 (Future)	PRODUCT WATER TREATMENT AND PUMPING	060
WWW12	Wet Weather Treatment Facility 1 Chemical Area	Product Water Post-Treatment	061
WWW22	Wet Weather Treatment Facility 2 Chemical Area (Future)	Product Water Storage	062
WWW13	Wet Weather Treatment Facility 1 Sludge Storage	Product Water Pumping	063
WWW23	Wet Weather Treatment Facility 2 Sludge Storage (Future)	BIOSEDIDS TREATMENT	100
WSW00	Site Overall	Primary Sludge Thickening	101
WSW01	PUB/NEA Administration, Visitor Centre	Damper	102
WSW02	Operation & Maintenance Building	Data Outlet (R145)	DOUT
WSW13	Guard House 1	PS Thickening Polymer System	103
WSW23	Guard House 2	WAS Pre-dewatering Polymer System	104
WSW04	Fire Water	Dewatering	105
WSW05	Near Shore Outfall	Dewatering Polymer System	106
WSW06	Chemical Unloading Area	BIOSEDIDS DIGESTION	110
WSW07	Tuas Port Local Sewer	Digestion Pretreatment	111
WSW08	Main Distribution Frame (MDF) Room	Digestion Feed Systems	112
WSW30	C2A-Sitewide	Digestion	113
WSW31	C3A-Sitewide	Digestion Scum Removal	114
WSW32	C3B-Sitewide	Biogas System	115
WSW33	C3C-Sitewide	Food Waste Processing	116
WSW34	C2B-Sitewide	GREASY WASTE PROCESSING	120
WSW35	C2B-Sitewide (NOT USED) Previously - Product Pumping	Greasy Waste Receiving	121
WSW36	C4A-Sitewide	Greasy Waste Screening	122
WSW37	C5A-Sitewide	Greasy Waste Transfer Pumping	125
WSW38	C6A-Sitewide	SIDESTREAM TREATMENT	200
WSW39	C6B-Sitewide	Sidestream N Solids Separation	201
WSW40	C6C-Sitewide	Sidestream N Bioreactor	202
WSW41	C4B Sitewide	Sidestream N Blowers	203
WSW42	C3E Sitewide	Sidestream N Wasting and Separation	204
WSW43	C3D1 Sitewide	Sidestream P Removal	205
WSW44	C3D2 Sitewide	WET WEATHER TREATMENT	300
WSW45	C3B1 Sitewide	WET WEATHER TREATMENT	301
WSW46	C3B2 Sitewide	CHEMICAL SYSTEMS	400
WSE00	Electrical Common	Ferric Chloride Storage and Feed	401
WSE11	Electrical Main Substation A	Sodium Hypochlorite Storage and Feed	402
WSE21	Electrical Main Substation B	Sodium Hydroxide Storage and Feed	403
WWM00	IWMF Common	Sulphuric Acid Storage and Feed	404
WGS00	General Services	Citric Acid Storage and Feed	405
WGS01	Site Completion	Sodium Bisulphite Storage and Feed	406
WTF00	Temporary Common	Ammonia Storage and Feed	407
WTF01	22kV Substation (for Construction)	Lime Storage and Feed	408
WTF02	Barging Facility	ODOUR CONTROL	500
WNP00	NEWater Industrial Water Pipeline Common	OCF Fans	501
WNP01	NEWater Industrial Water Pipeline Shaft	OCF Biotrickling Filters	502
WNS00	NEWater Common	OCF Chemical Scrubbers	503
WNS01	NEWater Storage	OCF Activated Carbon	504
WNS02	IW Storage	OCF Stack	505
WNS03	NW/IW Pump Station	OCF Biofilters	506
WNS04	NW/IW Post Treatment Chemical		
WNS05	NW/IW Post Treatment Lime		
WTX08	Tunnel Contract 08		

EQUIPMENT AND VALVE TYPE CODE TABLE

Process and Non-Process Equipment		Process and Non-Process Equipment		Process and Non-Process Equipment	
Process and Non-Process Equipment	DDDD	Process and Non-Process Equipment	DDDD	Process and Non-Process Equipment	DDDD
Accumulator	ACC	Flow Indicator	FI	Public Address Speakers	PAS
Active Harmonic Filter	AHF	Flow Switch High	FSH	Pull Box	PBX
Air Conditioning Unit	ACU	Flow Switch Low	FSL	Pulsation Dampener	AHP
Air Cooler	AC	Flow Transmitter	FIT	Pump	P
Air Dryer	ADR	Foam Separator	ROP	Relief Damper	ALS
Air Handling Unit	AHU	Gas Flare	GF	Remote I/O Panel	RIO
Air Receiver	AR	Gas Holder	GH	Remote Terminal Unit	RTU
Alarm Processing Unit (FIDS)	APU	Gas Infrared Heater	GIH	Router	PRO
Analysar Panel	ACP	Gas Unit Heater	GUH	Router (Non-Process)	ROUN
Analysar Sensor	AE	Generator	GEN	Router (process)	ROUP
Analytical Indicator Transmitter	AIT	Generator - Diesel	SDE	Rupture Disk	RD
Asset Management System Tag	AMST	Generator - Dual Fuel	DGE	Safety Shower	SWR
Automatic Transfer Control	ATC	Generator - LPG/Gas	SA	Sampler (Liquid)	CDG
Automatic Transfer Switch	ATS	Generator Control Panel	GCP	Sampler (Sludge)	SAS
Auxiliary Relay Panel	ARP	Gravity Belt Thickener	GBT	SCADA Client Work Station	SCWS
Backflow Preventer	BFP	Grit Classifier	GRC	Scraper	SCP
Ballast Recovery System	BRC	Grit Washer	GRW	Screen	CL
Battery	BAT	Hand Damper	HDP	Screenings Washer	SW
Battery Charger	BC	Heat Exchanger	HTX	Scrubber	SCU
Bin	CB	Heater	HTR	Scum Collector	SC
Blower	B	High Voltage Switchgear	HVSG	Security Access Card Reader	ACCR
Boiler	BLR	Hopper	HO	Security Access Electro Magnetic Lock	AEML
Boiling Access Controller	BACC	Hose Reel	HR	Security Access Reader (Biometric)	ACRB
Building Management System Field Level Controller	BMFC	Human machine Interface/Operator Interface Panel	OIP	Sediment Trap	TRP
Building Management System Plant Controller	BMPC	I/O Server	IOSV	Server (Non-Process)	SVRN
Building Management System Server	BMSS	Instrument Control Panel (Field)	ICP	Server (Process)	SVRP
Building Management System Workstation	BMWS	IP Microphone	IPMP	Server Rack	SVRR
Building Managment System Controller (DDC)	BMSC	IP Network Audio Adaptor	IPNA	Silencer	IS
Calibration Cylinder	CC	IP Network Switch	IPNS	Silo	SIL
Camera (ISMS Security)	SCAM	IP Phone Handset	IPH	Siren	SRN
Camera (Plant CCTV)	PCAM	IP Private Branch Exchange (PBX)	IPBX	Slide Gate - Actuated	SGM
Camera (Process Control)	CCAM	IP Telephony Server	IPTS	Slide Gate - Hand	SGH
Catalyst	CAT	Isolator	ISO	Soft Starter	SS
Catchpot	CPT	Junction Box	JB	Softener	SFT
CD/MP3 Player	CDMP	Lamella Plates	LPT	Static Mixer	SMX
Centrifuge - Dewatering	DWC	Level Element	LE	Stationary Louvre	SL
Centrifuge - Predewatering	PDC	Level Gauge	LG	Stop Log	SPL
Centrifuge - Thickening	TC	Level Indicator Transmitter	LIT	Strainer	STR
Chiller - Water	CH	Level Switch High	LSH	Sub Distribution Board	SDB
Coarse Bubble Diffuser	CBD	Level Switch Low	LSL	Sub Fire Alarm Panel	SFAP
Communication Rack	COMR	Lift Car	LC	Suction Diffuser	SD
Compressor - Air	CA	Lighting & Small Power Distribution Board	LDB	Tank	TNK
Compressor - Gas	CG	Local Control Panel	LCP	Telecom Consumer Distribution Panel	TDPC
Compressor Condensing Unit	CCU	Local Control Station	LCS	Telecom Distribution Panel (by Service Provider)	TDPS
Condensate Trap	CT	Local Indicator Board	LIB	Telephone Application Server	TAS
Conveyor	CNV	Low Voltage Switchgear	LVSG	Temperature Element	TE
Core Distribution Network Switch	CDNS	Macerator	MCR	Temperature Indicator Transmitter	TIT
Crane - Trolley	CRN	Main Distribution Board	MDB	Temperature Switch High	TSH
Cyber Security Data Diode	DDO	Main Distribution Frame	DMF	Temperature Switch Low	TSL
Cyber Security Intrusion Detection System	IDS	Main Fire Alarm Panel	MFAP	Torque Switch High	NSH
Cyber Security Intrusion Protection System	IPS	Makeup Air Unit	MAU	Transformer	TRF
Cyclone	CYC	Manhole	MH	Turbine	TUB
Damper	DMP	Marshaling Panel	MRSP	Uninterruptible Power Supply	UPS
Data Outlet (R145)	DOUT	MCSA (Motor Current Signature Analysis) Analyser Module	MAM	Uninterruptible Power Supply Distribution Board	UPSDB
DC Distribution Board	DCDB	Mechanical Condition Module (Vibration Monitoring)	MCM	UPS Battery Rack	BATR
Density Indicating Transmitter	DIR	Media Converter (PoE enable)	PMC	UV Reactor	UVR
Diffuser	DF	Media Converter Chassis	PMCC	Valve - Air Release (combination)	ARV
Direct On-line Starter	DOL	Medium Voltage Switchgear	MVSG	Valve - Air Release (in)	ARVI
Diverter Chute	CHU	Membrane	MB	Valve - Air Release (out)	ARVO
Domain Controller	DCTR	Miscellaneous	MSC	Valve - Back Pressure	BVP
Door Contact	DRC	Mixer	MRX	Valve - Check	NRV
Door Release	DRR	Motion Switch High	MSH	Valve - Electric Actuated	MV
Door Release Breakglass	DRBG	Motion Switch Low	MSL	Valve - Emergency Shutdown	ESD
Dosing Skid	DS	Motor	MTR	Valve - Float Actuated	LRV
Edge Switch	IPES	Motor Control Centre	MCC	Valve - Flow Control	FCV
Eductor	E	Motorised Damper	MDP	Valve - Hand	HV
Electrical Interface Cabinet (PLC Panel)	EICP	Network Switch (Edge Switch- Non Process)	ESWN	Valve - Level Control	LCV
Essential Services Switchboard	ESSB	Network Switch (Main Backbone - Non Process)	NSWN	Valve - Pneumatic Actuated	AV
Ethernet Media Converter	PEMC	Network Switch (Main Backbone - Process Automation Network)	NSWPA	Valve - Pressure Control	PCV
F/O to Copper Media Inverter	FCMC	Network Switch (Main Backbone - Process Monitoring Network)	NSWPM	Valve - Pressure Regulating	PRV
Fan - Exhaust	FEU	Network Switch (Sub-Ring Switch - Process)	SSWP	Valve - Pressure Relief	SRV
Fan - Supply	FSU	Neutral Earthing Resistor	NER	Valve - Safety Selector	SSV
Fibre Optic Breakout Tray (non-process)	FBTN	PA Amplifiers	PAA	Valve - Solenoid Actuated	SV
Fibre Optic Breakout Tray (Process)	FBIT	Package Unit	PU	Valve - Sprinkler Control	SCV
Fibre Optic Interface Panel (Non-Process)	FOIN	Penstock - Hand	PSK	Valve - Temperature Control	TCV
Fibre Optic Interface Panel (Process)	FOIP	Penstock - Motorised	PSM	Valve - Thermal Shutdown	TSV
Field Ethernet Switch	FSWP	Personnel Tracking Device	PTD	Valve - Vacuum Pressure Relief Valve	SVRV
Field Fibre Optic Termination Panel	FFOTP	Personnel Tracking System Access Point	PTAP	Valve - Vacuum Relief	VRV
Field Junction Box (Non-process)	FJBN	PLC Panel (Package Equipment)	ECCP	Vaporizer	VAP
Field Junction Box (process)	FJBP	Power Factor Correction Capacitor	PFCC	Variable Frequency Drive	VFD
Filter	FTR	Power/Energy Monitoring Device	PMD	Vibration Monitoring System	VMS
Fine Bubble Diffuser	FBA	Pressure Differential Indicator Transmitter	PDIT	Vibration Sensor	VT
Fire Hose Reel	FHR	Pressure Differential Switch	PDS	Video Recording Manager (ISMS Security)	SVRM
Fire Hydrant	FH	Pressure Element	PE	Video Recording Manager (Plant CCTV)	PVRM
Fireman Intercom Handset	FIHS	Pressure Indicator	PI	Video Wall	VDW
Fireman Intercom Main Panel	FIMP	Pressure Indicator Transmitter	PIT	Volumetric Chemical Feeder	VCF
Firewall (Non-process)	FRWN	Pressure Switch High	PSH	Weight Indicator Transmitter	WIT
Firewall (Process)	FRWP	Pressure Switch Low	PSL	Weighted Damper	WDP
Flame Arrestor	FA	Pressure Vessel	PV	Wireless Acess Point	WAP
Flow Control Gate	FCG	Printer	PRNT	Work Station (non-process)	WSNP
Flow Control Penstock	FCP	Programmable Logic Controller	PLC	Work Station (Process)	WSP
Flow Element	FE			Workstation	PWS

FLUIDS CODE TABLE

Fluid Streams		Fluid Streams	
Fluid Streams	CCCC	Fluid Streams	CCCC
Activated Carbon	AC	Instrument Air	IA
Aeration Air	AA	Instrumentation	I
Air High Pressure	AHP	Lime Hydrated	LH
Air Low Pressure	ALP	Lime Sludge	LSL
Aluminium Sulphate	ALS	Lime Solution	LS
Ammonium Sulphate	NHS	Liquid Oxygen	LOX
Antiscalant	ASC	Low Pressure Hot Water	LPWH
Backwash Supply	BWS	Low Voltage	LV
Backwash Waste	BWW	Lubricating Oil	LO
Ballast	BAL	Magnesium Hydroxide	MH
Biomass Nutrient	BN	Membrane Filtrate	MF
BTF Liquor	BL	Mixed Liquor	ML
Carbon Dioxide (Gas)	CDG	Mixed Liquor Return	MCR
Carbon Dioxide (Liquid)	CDL	Monochloramine	NCA
Chilled Water Return	CHWR	NEWater	NW
Chilled Water Supply	CHWS	Nitrogen	NP
Chlorine	CL	Non-Process Communication	OGW
CIP Return	CIPR	Organic Waste	OGW
CIP Supply	CIPS	Oxygen (Gas)	PD
Citric Acid	CIT	Plant Drain	P2
Clarified Water	CLW	Plant Effluent	P1E
Combustion/Pre-heated Air	CP	Plant Water (DLM MF)	G2D
Concentrated Greasy Waste	CGW	Plant Water (ILM MF)	G2I
Concentrated Scum	CSM	Plant Water (ROP)	G1
Condensate	SC	Plant Water High Pressure	G2HP
Control	C	Polyelectrolyte	PO
Cooling Liquid (Glycol, Brine, etc.)	KL	Potable Water	PW
Cooling Water Return	CWR	Primary Effluent	PE
Cooling Water Send	CWS	Primary Influent	PI
Deminerlized water	DM	Primary Scum	PSM
Dewatered Sludge	DWS	Primary Sludge Domestic	PSD
Dewatered Waste Activated Sludge	DWAS	Primary Sludge Industrial	PSI
Dewatering Centrate	DC	Return Activated Sludge	RAS
Dewatering Centrate (WAS Pre-dewatering)	DCW	Return Primary Sludge	RPS
Digested Sludge	DS	RO Brine	ROBR
Domestic Used Water	DUW	RO Concentrate	ROC
Electricity	E	RO Feed	ROF
Exhaust Air	EA	RO Permeate	ROP
Ferric Chloride	FC	Sample	S
Ferric Sulphate	FS	Scour Air	AS
Fire Hydrant Water	FH	Screenings	SG
Fire Protection Foam	FPE	Scrubber Recycle	SR
Fire Water	FPW	Secondary Scum	SSM
Flue Gas	FU	Sludge Return Liquor	SRL
Food Waste	FW	Sodium Bisulphite	SBS
Foul (Sanitary) Drain	FD	Sodium Chloride	SCL
Foul Air	FA	Sodium Hydroxide	SHYD
Fuel Oil (Return)	FOR	Sodium Hypochlorite	SHYP
Fuel Oil (Supply)	POS	Softened Water	WS
Gas (Biogas)	GB	Steam High Pressure	STHP
Gas (Mains)	GM	Steam Low Pressure	STLP
Greasy Waste	GW	Sulphuric Acid	SA
Grit	GR	Surface Water Drain	WD
Grit Slurry	GRS	Telecoms	T
High Pressure Hot Water	HPHW	Thickened Primary Sludge	TPS
High Voltage	HV	Thickened Waste Activated Sludge	TWAS
Hot Water (Return)	HWR	Thickening Centrate Domestic	TCD
Hot Water (Supply)	HWS	Thickening Centrate Industrial	TCI
Hydraulic Fluid Return	HFR	Treated Air	TA
Hydrochloric Acid	HA	Vacuum	VM
Hydrogen Peroxide	HP	Vapour Line	VA
Hydrolysed Sludge	HS	Waste Activated Sludge	WAS
Industrial Used Water	IUW	Waste Activated Sludge Domestic	WASD
Industrial Water	IW	Waste Activated Sludge Industrial	WASI
		Waste Chemical	WC
		Wet Weather Sludge	WWS

PIPE MATERIAL CODE TABLE

Pipe Materials	MMMM
Acrylonitrile Butadiene Styrene ABS	ABS
Carbon Steel Cement Lined	CSSL
Carbon Steel Fusion Bonded Epoxy Lined	CSEL
Carbon Steel Galvanised	CSG
Carbon Steel Polyurethane Lined	CSPL
Carbon Steel Unlined	CS
Copper	CU
Ductile Iron Cement Lined	DICL
Ductile Iron Ceramic Epoxy Lined	DICE
Ductile Iron Fusion Bonded Epoxy Lined	DIEL
Ductile Iron Glass Lined	DIGL
Ductile Iron Polyurethane Lined	DIPU
Ductile Iron Polyurethane Lined	DIPU
Ethylene Propylene Diene Monomer	EDPM
Fibreglass Reinforced Plastic	FRP
High Density Polyethylene	HDPE
Neoprene	CR
Nitrile Rubber	NBR
Polyethylene	PE
Polypropylene	PP
Polyvinyl Chloride Chlorinated	CPVC
Polyvinyl Chloride Unplasticised	UPVC
Polyvinylidene Fluoride	PVDF
Reinforced Concrete Pipe (HDPE Lined)	RCP
Stainless Steel 304L	SS04
Stainless Steel 316L	SS16
Vitrified Clay	VCP

TENDER

T1	20/05/22	ISSUED FOR TENDER C4B	CN	GOT
T1	02/07/21	ISSUED FOR TENDER C3D2	CN	GOT
T1	18/12/20	ISSUED FOR TENDER C3B1	CN	GOT
T2	06/10/20	ISSUED FOR ADDENDUM 2 C2B	CN	GOT
T1	23/07/20	ISSUED FOR TENDER C2B	CN	GOT
T1	24/04/20	ISSUED FOR TENDER C3A	CN	GOT
REV	DATE	DESCRIPTION	REVIEWED	AUTHORISED

CLIENT: